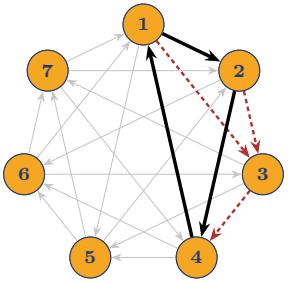
(a) the tournament P_7



$T|_B$, $B = \{1,2,4\}$
arcs at 3

(b) the base state, forced

	0	1	2	3
π_1	1	2	4	
π_2	2	4	1	
π_3	4	1	2	

the only one

options, as slot triples:

$D(3)$ 113 203 212 213 230
$|D(3)| = 5$ fewest, so $v^* = 3$

$D(5)$ 032 122 131 132 302
$|D(5)| = 5$

$D(6)$ 023 221 311 320 321
$|D(6)| = 5$

$D(7)$ 000 001 002 003 010 011
012 020 030 100 101 110
111 120 200 201 300
$|D(7)| = 17$

(c) two rejected insertions of 3

put 3 at 113:

π_1	1	3	2	4
π_2	2	3	4	1
π_3	4	1	2	3

× breaks Corollary 2.2: no voter can still lead with $(1,2)$ or $(1,4)$

put 3 at 213:

π_1	1	2	3	4
π_2	2	3	4	1
π_3	4	1	2	3

× $D(5 \mid B \cup \{3\}) = \varnothing$; and it makes $2 \to 3$ unanimous

(d) an insertion of 3 that survives

put 3 at 230:

π_1	1	2	3	4
π_2	2	4	1	3
π_3	3	4	1	2

✓ π_1 still leads with $(1,2)$, and nothing has run out of room

what that leaves of (b)'s options:

$D(5)$ 043 142 143 403
$|D(5)| = 4$ was 5

$D(6)$ 024 222 420
$|D(6)| = 3$ was 5

$D(7)$ 041 301 302 401
$|D(7)| = 4$ was 17

(e) refining $D(6)$

past 230:

tuple	*per voter*		*lift*	$c_6(3)$	
023	=, =, +1	→	024	2	✓
221	**split**, =, +1	→	222	2	✓
		→	~~322~~	1	×
311	+1, =, +1	→	~~412~~	1	×
320	+1, =, **split**	→	420	2	✓
		→	~~421~~	1	×
321	+1, =, +1	→	~~422~~	1	×

(f) the completed profile

π	1	2	3	4	5	6	7
$g\pi$	5	7	2	4	6	1	3
$g^2\pi$	6	3	7	4	1	5	2

$g = (1\,5\,6)(2\,7\,3)$, of order 3
every arc at support 2